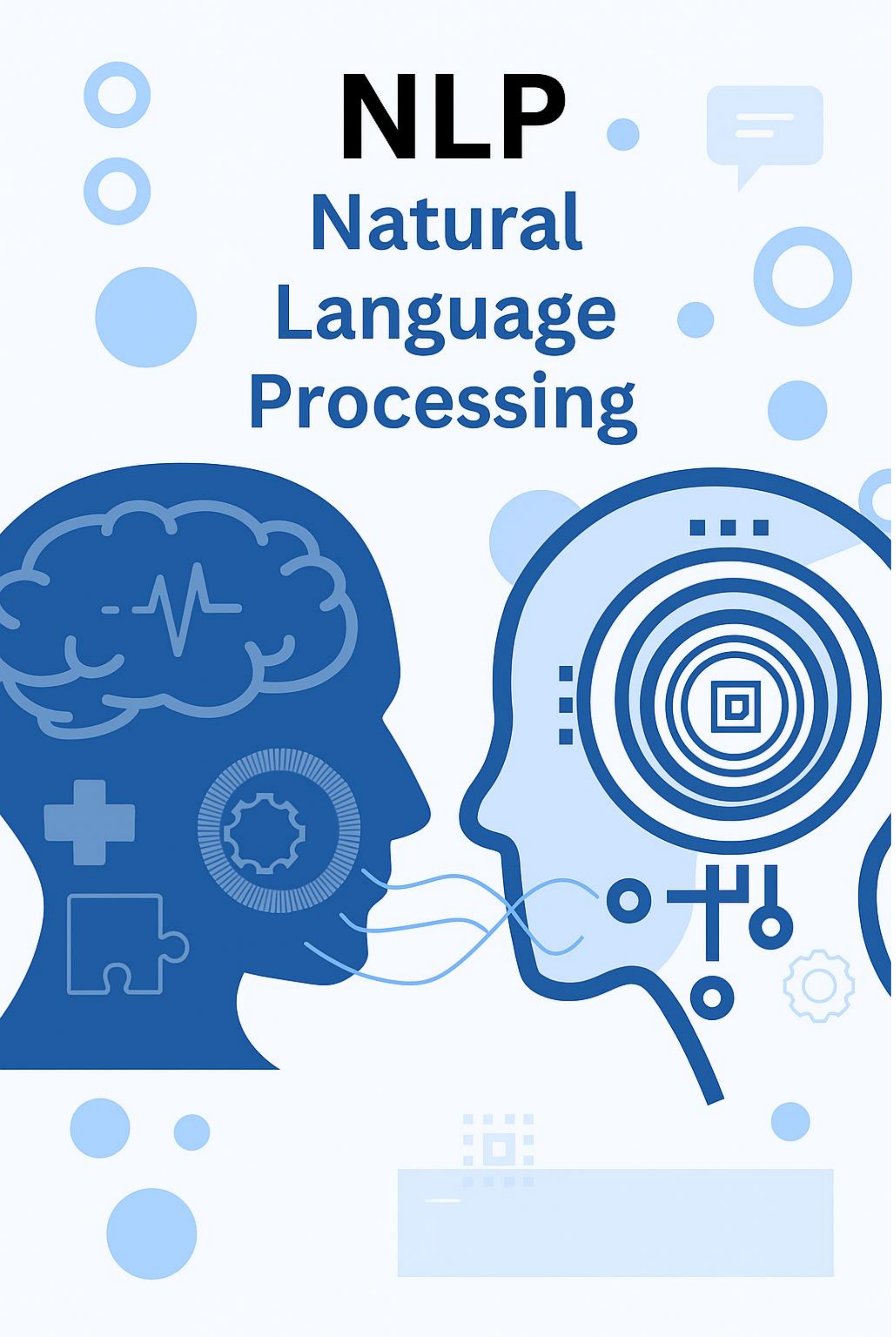


NATURAL LANGUAGE PROCESSING (NLP)

PMDS606L

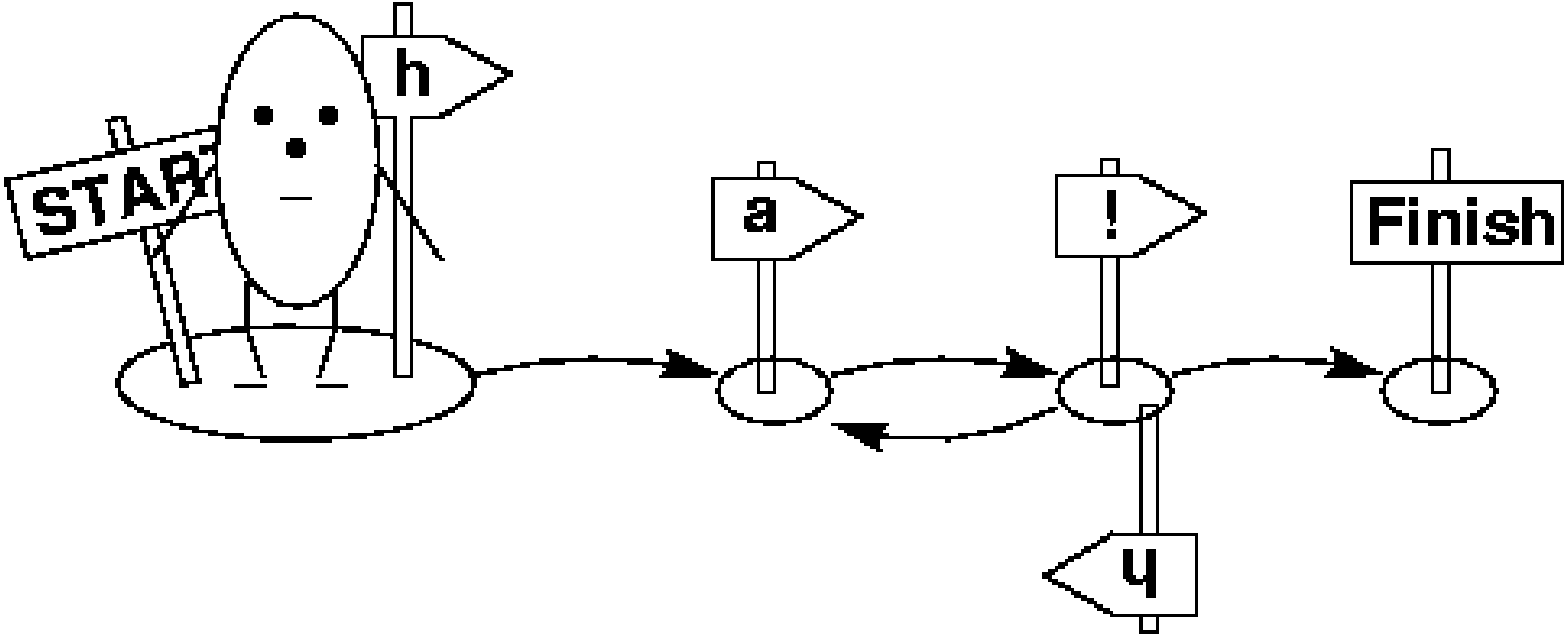
MODULE 4

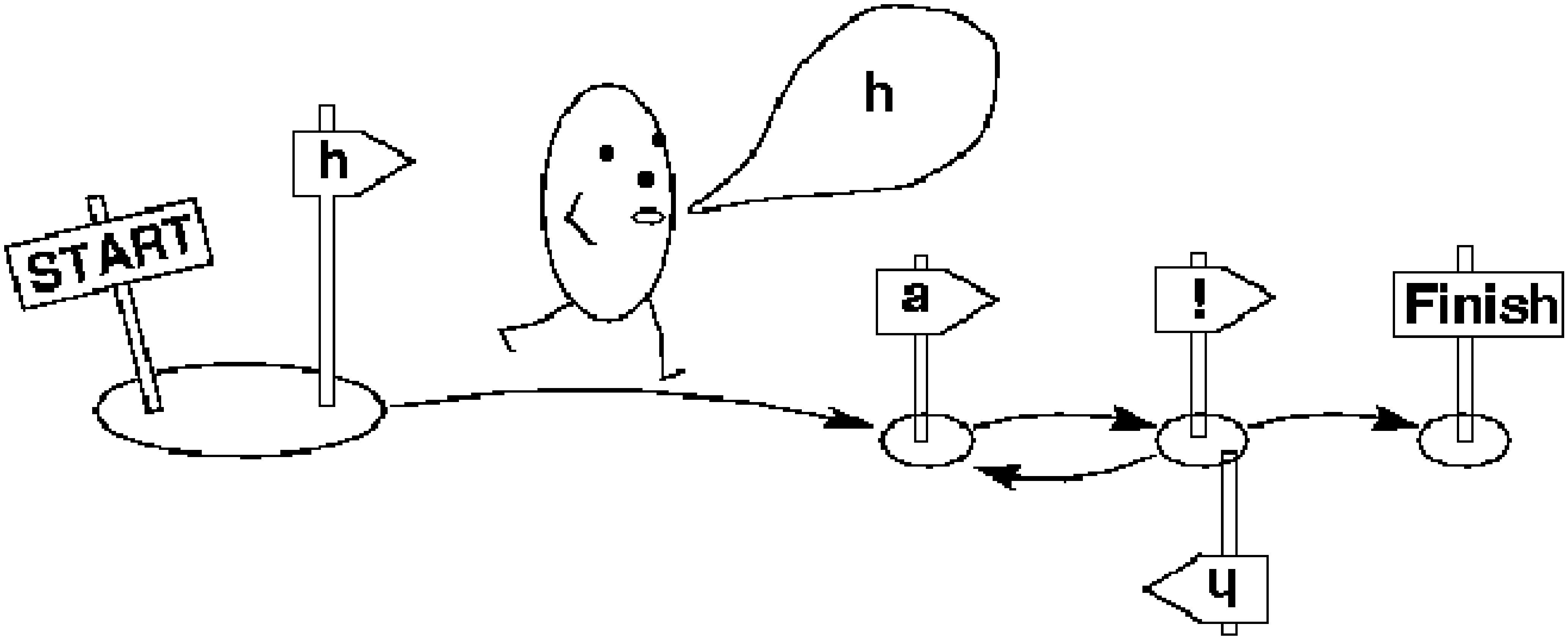
LECTURE 6

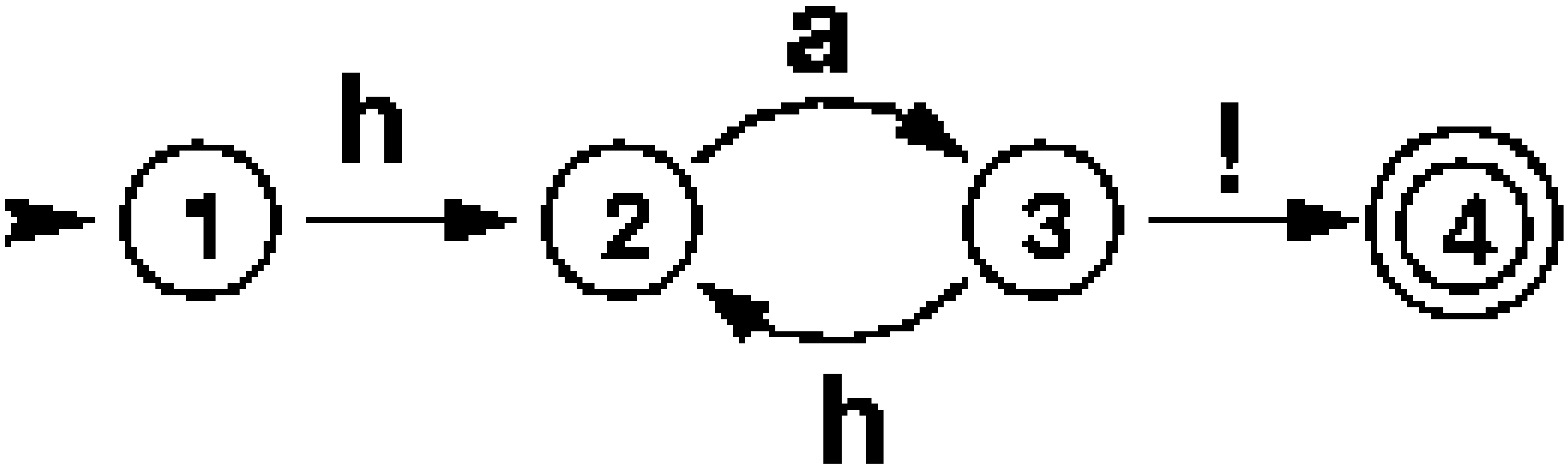
Dr. Kamanasish Bhattacharjee

Assistant Professor

Dept. of Analytics, SCOPE, VIT


STAR
h
a
!
Finish


START
h
h
a
!
y
Finish



h
a
!
h
1
2
3
4

FSA and FST

- **FSA – Finite State Automata**: It provides decisions regarding the acceptance and rejection of a string. It is useful in deciding whether a given word belongs to a particular language or not.

- **FST – Finite State Transducers**: FST generates output for a given input, i.e., maps input-output strings.

DETERMINISTIC FINITE AUTOMATA (DFA)

Mathematically, a DFA can be represented by 5-tuple (Q, Σ, δ, q0, F), where –

- *Q is a finite set of states.*
- *Σ is a finite set of symbols, called the alphabet of the automaton.*
- *δ is the transition function where δ: $Q \times \Sigma \rightarrow Q$.*
- *q0 is the initial state from where any input is processed ($q0 \in Q$).*
- *F is a set of final state/states of Q ($F \subseteq Q$).*

Whereas graphically, a DFA can be represented by diagraphs called state diagrams where –

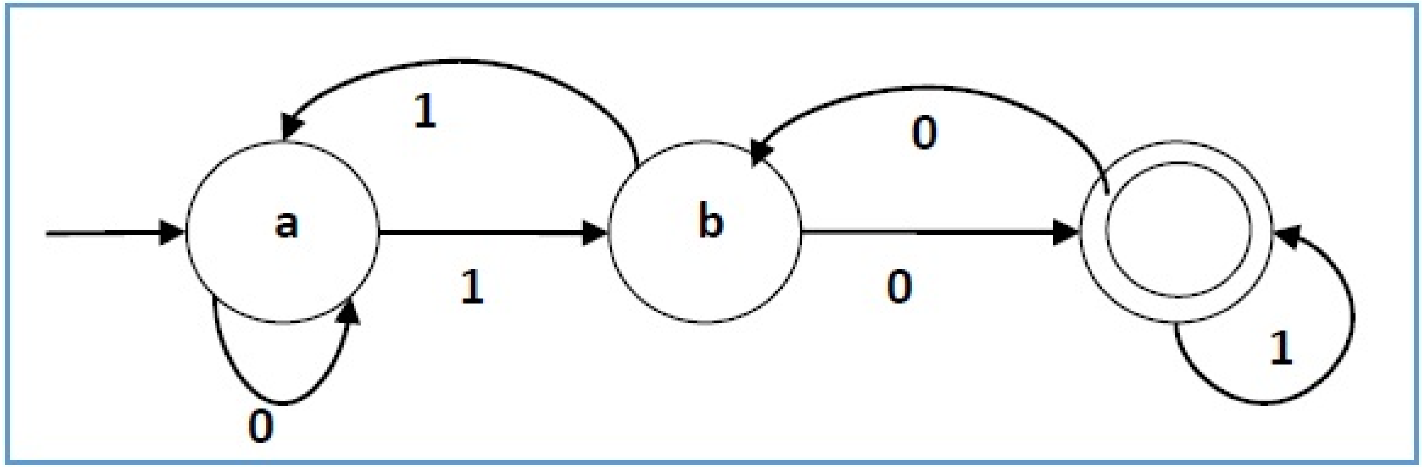
- *The states are represented by vertices.*
- *The transitions are shown by labeled arcs.*
- *The initial state is represented by an empty incoming arc.*
- *The final state is represented by double circle.*

DETERMINISTIC FINITE AUTOMATA (DFA)

Transition function δ as shown by the following table –

Present State	Next State for Input 0	Next State for Input 1
a	a	b
b	c	a
c	b	c

Its graphical representation would be as follows –



NON- DETERMINISTIC FINITE AUTOMATA (NDFA)

Mathematically, NDFA can be represented by 5-tuple (Q, Σ, δ, q0, F), where −

- *Q is a finite set of states.*
- *Σ is a finite set of symbols, called the alphabet of the automaton.*
- *δ is the transition function where δ: $Q \times \Sigma \rightarrow 2\ Q$.*
- *q0 is the initial state from where any input is processed ($q0 \in Q$).*
- *F is a set of final state/states of Q ($F \subseteq Q$).*

Whereas graphically (same as DFA), a NDFA can be represented by diagraphs called state diagrams where −

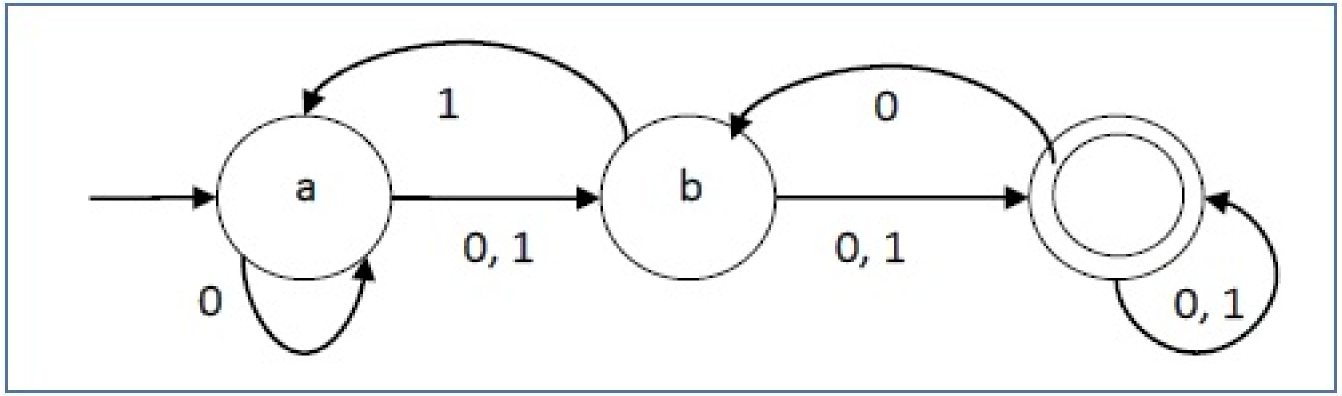
- *The states are represented by vertices.*
- *The transitions are shown by labeled arcs.*
- *The initial state is represented by an empty incoming arc.*
- *The final state is represented by double circle.*

NON- DETERMINISTIC FINITE AUTOMATA (NDFA)

The transition function δ as shown below –

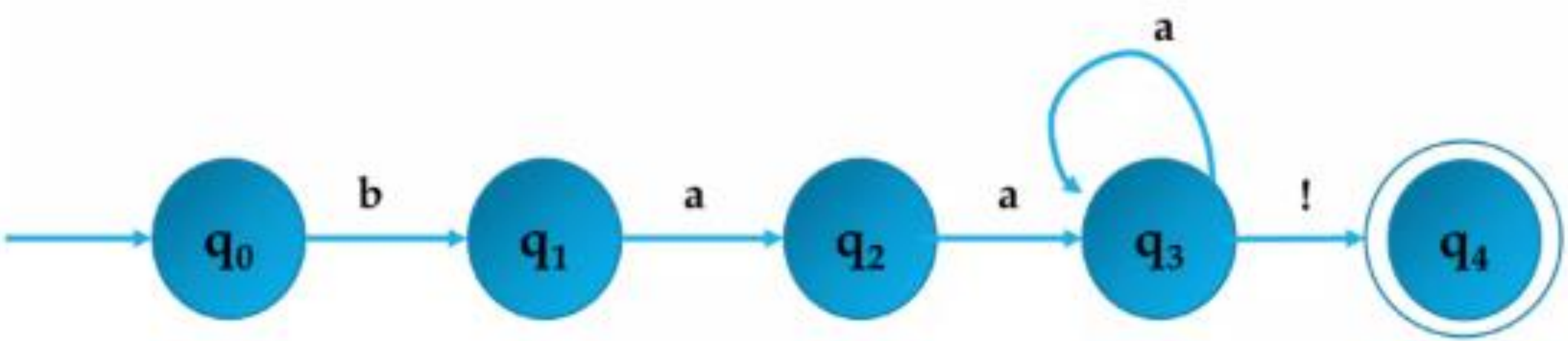
Present State	Next State for Input 0	Next State for Input 1
a	a, b	b
b	c	a, c
c	b, c	c

Its graphical representation would be as follows –



FSA EAMPLE

Exercise: Design a Finite State Automata for baa+!



Q: Finite set of states = q0, q1, q2, q3, q4

$\sum$: Set of input alphabets = {a,b,!}

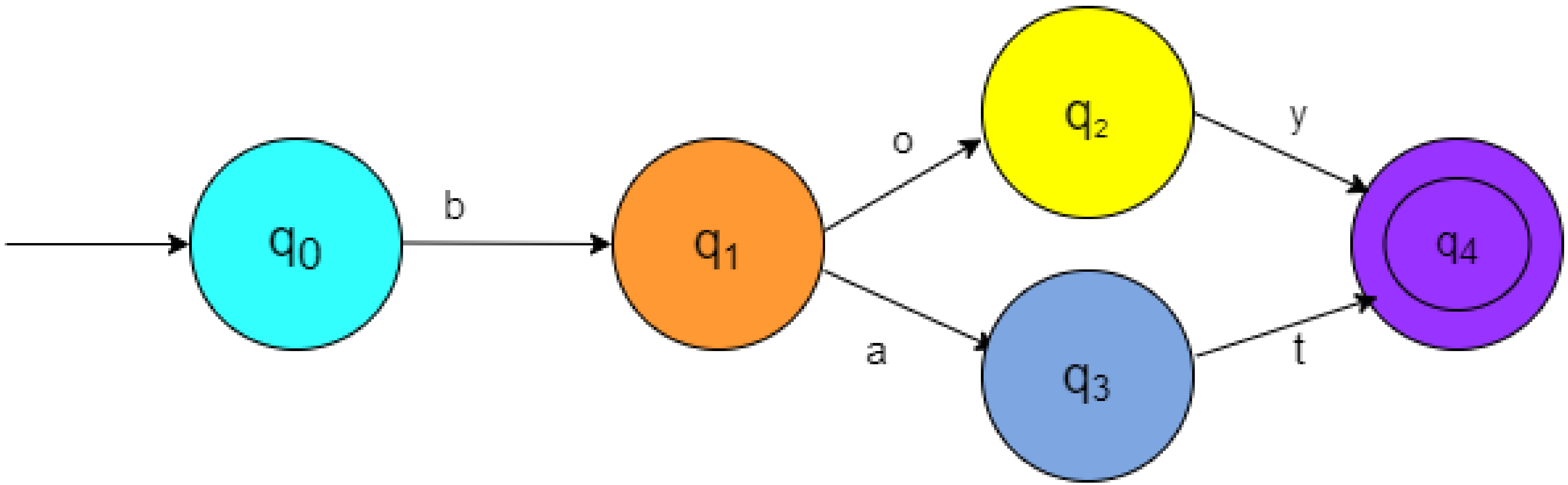
q0: Start state

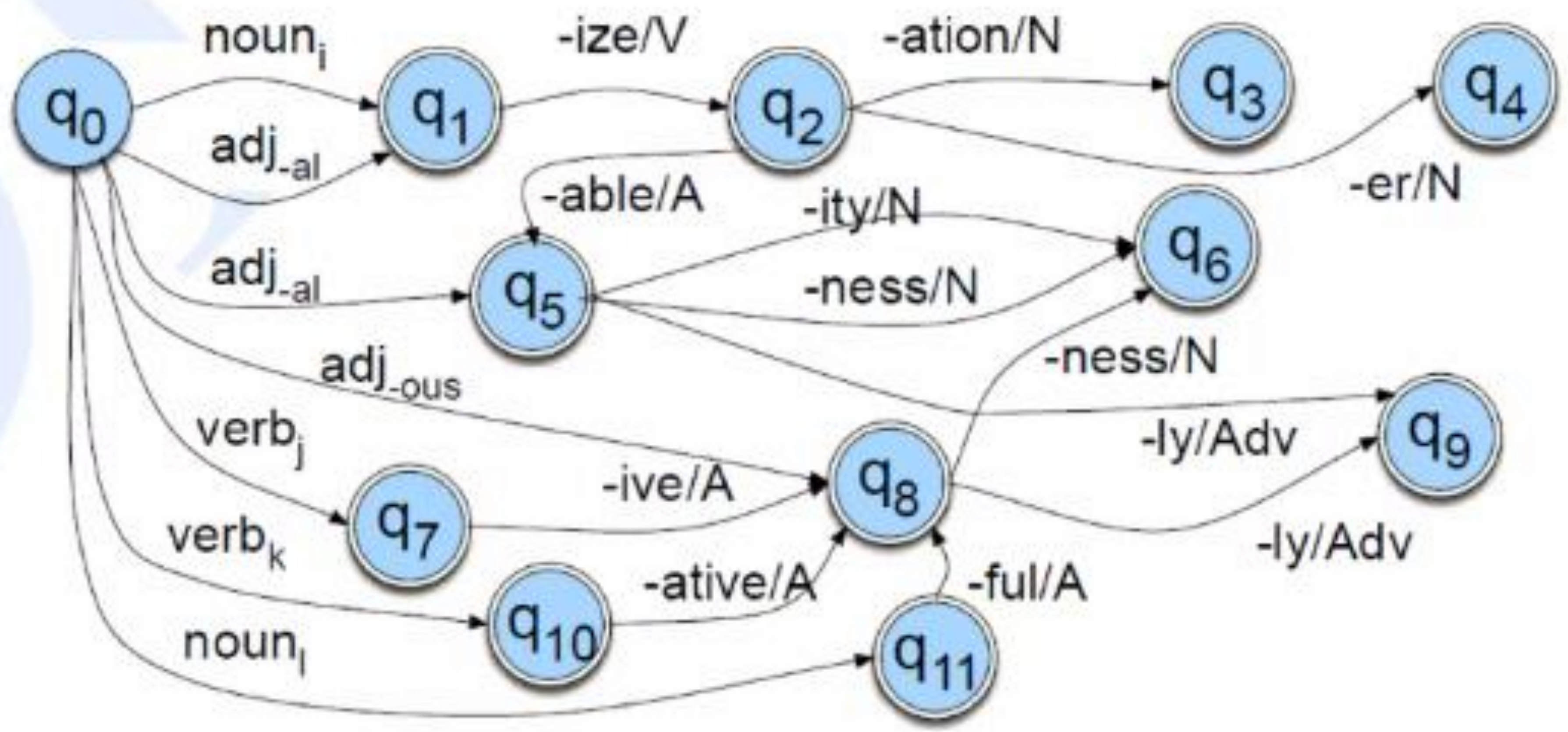
F: Set of final states {q4}

δ(q, i) defined by the transition table

FSA EXAMPLE

	a	b	!
q_0	ϕ	q_1	ϕ
q_1	q_2	ϕ	ϕ
q_2	q_3	ϕ	ϕ
q_3	q_3	ϕ	q_4
q_4	ϕ	ϕ	ϕ


q0
b
q1
o
a
q2
q3
y
t
q4



An FSA for another fragment of English derivational morphology

This FSA models several derivational facts, such as the well-known generalization that any verb ending in *-ize* can be followed by the nominalizing suffix *-ation*. Thus, since there is a word *fossilize*, we can predict the word *fossilization* by following states *q*0, *q*1, and *q*2. Similarly, adjectives ending in *-al* or *-able* at *q*5 (*equal*, *formal*, *realizable*) can take the suffix *-ity*, or sometimes the suffix *-ness* to state *q*6 (*naturalness*, *casualness*)